

Topic 5 of Personnel Economics: Selection and Training

Exercises

Exercise 5.1

A stylish restaurant is in need of waiters. Two types of waiters can be hired: people with and people without a diploma. Waiters with a diploma are much quicker in giving advice and in serving dinners. If the firm hires waiters with a diploma, it needs two waiters, each of whom will yield sales of 500 euro per evening. If the firm hires waiters without a diploma, it needs three waiters, each yielding 300 euro per evening.

a) Suppose a waiter with a diploma costs 100 euro per evening, whereas a waiter without a diploma costs 25 euro per evening. Which type of worker will be hired by a profit-maximizing restaurant?

The restaurant wants its waiters to be properly dressed. Suppose the dry-cleaning of working cloth cost the restaurant 30 euro's per worker per evening.

b) Which type of worker will be hired by a profit-maximizing restaurant, taking into account the cost of dry-cleaning working cloth?

c) At what level of cost of dry-cleaning is the restaurant indifferent between the two types of workers?

Exercise 5.2

Consider the following example. All amounts are in euros per year. The firm wants to maximize its profits.

	Production value	Wage
Low-skilled worker	50.000	15.000
High-skilled worker	80.000	35.000

a) Suppose the firm has very limited space so that it can only hire one worker. Which type of worker will the firm hire?

b) Suppose the firm can, at a yearly rate of 10.000 euros, rent additional space so that it can employ two workers. Will the firm rent additional space? Which type of worker will the firm hire?

c) Suppose the firm has unlimited space, but market demand is at most 800.000 euros. Which type of worker will it hire?

Exercise 5.3

A firm produces output using low-skilled workers. Production is given by:

$$Q = \theta \ln L$$

where L is number of low-skilled workers employed by the firm. The price of the product is P and wage of a low-skilled worker is w_l .

a) Derive the profit-maximizing number of low-skilled workers.

b) How does the profit-maximizing number of low-skilled workers depend on:

- the wage
- the price of the firm's product
- the technological parameter θ

Give interpretations for your results.

Exercise 5.4

A firm can choose between two types of workers: a proven one, who yields output 50 per period for sure, and a risky worker, who either yields 150 each period, which happens with probability 1/3, or yields -30 each period, which happens with probability 2/3. Assume that a worker's wage is independent of type.

a) Suppose the firm operates for one period. Which type of worker should the firm hire, if it aims to maximize profits?

b) Suppose the firm operates for two periods. It can hire a worker at the beginning of the first period as well as costlessly replace a worker at the beginning of the second period. Which type of worker should the firm hire at the beginning of the first period, if it aims to maximize profits?

c) Consider the same setting as in part b), but suppose now that there is a cost of replacing a worker at the beginning of the second period, denoted by c . Derive the value of c for which the firm is indifferent between hiring one or the other type of worker at the beginning of period 1.

Exercise 5.5

Suppose two types of workers exist, those with low ability and those with high ability. Workers know their own ability, but the firm can not observe a worker's type. Suppose getting a degree is more easy for a high-ability worker than for a low-ability worker. The monetary equivalent of the utility loss from obtaining a degree is 30 for a high-ability worker and 70 for a low-ability worker.

a) Suppose the market life-time wage for workers with a degree is 200, and for workers without a degree 100. Will having a degree reveal a worker's ability to the firm?

b) And when wages are 150 and 100 respectively?

Exercise 5.6

Suppose a firm can sell 900 shoes at a price of 3. To produce the shoes, the firm can choose between two types of workers: energetic workers and lazy workers. The hourly output of the different types of workers and their outside hourly wage are given in the table:

	Output p/h	Outside wage p/h
Energetic	9	18
Lazy	5	12

a) Suppose the firm can observe a worker's type. Which type does the firm prefer if it strives for maximum profits?

Suppose now that the firm can not readily observe whether a worker is energetic or lazy.

b) At what level of piece-rate pay can the firm be sure it attracts only its most preferred type of worker?

c) Suppose the trade unions impose a minimum piece rate of 2,50. Is there any way the firm can still ensure that only its most preferred type of worker apply?

Exercise 5.7

A firm wants to attract skilled workers, but can not observe whether workers are skilled or unskilled. Workers do know their skill level. Let:

W_s = skilled worker's alternative per-period wage

W_u = unskilled worker's alternative per-period wage

W_1 = wage during probationary period

W_2 = post-probationary wage

P_s = probability that a skilled worker passes probation.

P_u = probability that an unskilled worker passes probation.

Assume that probationary and post-probationary period are equal in length.

Assume $P_u=0$. Derive the levels of W_1 and W_2 which guarantee that only skilled workers have an incentive to apply at this firm and interpret your results.

Exercise 5.8

Consider an economy where firms compete for workers, and there is free entry and exit of firms. Suppose measuring a worker's output costs c euro per worker per day. Workers differ in their daily productivity. The least productive worker in the economy has productivity 0, the most productive has productivity 100. Workers are distributed uniformly over the interval 0 to 100. Firms can either pay a straight salary or offer piece-rate pay. In the latter case, output must be measured. None of the firms make profit.

- a) Suppose measurement cost are zero. Which workers will work at a salary firm?
- b) Suppose measurement cost are 10. Which workers will work at a salary firm?
- c) Derive the level of measurement cost at which all of the workers will work for straight salary.

Answers

5.1

- a) People without a diploma ($825 > 800$).
- b) People with a diploma ($740 > 735$).
- c) 25 euro.

5.2

- a) High-skilled worker
- b) Yes, it will rent additional space and hire an additional high-skilled worker.
- c) It will hire 16 low-skilled workers.

5.3

- a) $L = P \theta / w_l$
- b)
 - decreases with the wage
 - increases with the price of the firm's product
 - increases with the technological parameter θ

5.4

- a) A proven worker ($50 > 30$).
- b) A risky worker. Expected profits are $1/3 \times (150 + 150) + 2/3 \times (-30 + 50)$, which is higher than $50 + 50$. (Use backward induction).
- c) $c = 20$. ($100 = 1/3 \times (150 + 150) + 2/3 \times (-30 - c + 50)$ gives $c = 20$).

5.5

- a) No, all will obtain a degree.
- b) Yes, only the high-ability workers will get a degree.

5.6

- a) Energetic cost $18/9 = 2$ per unit of output; lazy workers are more expensive per unit of output.
- b) Piece-rate should be higher than or equal to 2 and lower than 2,40.
- c) Yes, but only if the firm can ask a fee. A fee of a little over 0,50 works. It implies that lazy workers can make $5 \times 2,50 - 0,50 < 12$, whereas energetic workers make $9 \times 2,50 - 0,50 = 22 > 18$. Clearly, the fee should not be higher than 4,50.

5.7

$W_1 < W_u$ and $W_2 > [1/P_s][(1 + P_s)W_s - W_1]$.

5.8

- a) None. The counterfactual is instructive here. Suppose workers with productivity 0-40 work for straight pay. Zero-profit condition implies that pay will be 20 (the average productivity). This gives all workers with productivity higher than 20 an incentive to work for a piece-rate firm. And so on...
- b) All workers with productivity less than 20. Define q^* as the productivity of the indifferent worker, with all workers having productivity equal to or less than q^* willing to work at a

straight-pay firm. Due to the zero-profit condition, workers in straight-salary firm earn $q^*/2$.
Under piece-rate: $q-10$. Solving $q/2 = q - 10$ gives $q=20$.
c) $c=50$.